Robert Burdsall

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Summary

Electrical Engineering student at NC State with industry experience building manufacturing automation end to end: PLC programming, control panel design, and machine vision integration. Led the design of an inspection machine running 120 parts/min from concept through commissioning, and brought first-time automated defect detection to 3 production machines. Seeking a controls or automation engineering internship in robotics, autonomous systems, or manufacturing.

Education

North Carolina State University — Raleigh, NC
Bachelor of Science in Electrical Engineering, GPA: 3.5 — *Expected Dec. 2028*

- **Relevant Coursework:** Introduction to Signals, Circuits and Systems; Fundamentals of Logic Design; Computer Systems Programming; Introduction to Computer Systems

North Carolina School of Science and Mathematics — Morganton, NC
High School Diploma, Computer Science Focus, GPA: 4.8 — *Aug. 2023 – May 2025*

Experience

IDEAL Fastener Corporation — Oxford, NC
Electrical Engineering Intern — *May 2026 – Aug. 2026*

- Led end-to-end design of a zipper cap inspection machine running 120 parts/min, from concept through commissioning, integrating Keyence machine vision, vacuum ejection, solenoid valve sequencing, a DOPSoft HMI, and PLC quality tracking with good/bad rate alarms.
- Programmed Omron PLCs in CX-Programmer, implementing ladder logic control sequences, interlocks, stepper motion pulse output, and buffered part tracking through the machine.
- Established automated defect detection for the first time on 3 existing production machines, reaching 100% inspection coverage by configuring Keyence IV3/IV4 vision systems and defining defect recognition criteria.
- Designed and fabricated 5+ electrical control panels, performing full wiring across relays, power supplies, DIN rail, disconnects, proximity and laser sensors, and solenoid valves.
- Produced TinyCAD schematics for every panel so plant technicians could replicate, service, and troubleshoot the hardware from documentation alone.

FIRST Robotics Competition, Team 9150 Labyrinth Robotics — Morganton, NC
Software Developer (2024–25); Lead Programming Mentor (2025–Present) — *Aug. 2024 – Present*

- Designed and implemented the full robot control software in Java using WPILib, integrating vision and dual scoring systems.
- Achieved the 3rd-highest scoring performance across 2025 North Carolina events, including the UNC Asheville District Event.
- Lead and mentor 25+ students, teaching Java, WPILib, wiring, and iterative test and debug of a competition robot.

North Carolina School of Science and Mathematics — Morganton, NC
Computer Science Teaching Assistant — *Aug. 2024 – May 2025*

- Taught 150+ students programming fundamentals in Java and Python.
- Built a cross-platform resource tracking application in Flutter and Dart in collaboration with faculty.

Projects

Aurum | *Dart, Flutter, Python, Express.js, TensorFlow* — Sep. 2024 – Dec. 2024

- Built a cross-platform mobile application for real-time coin appraisal, running on-device inference with TensorFlow Lite.
- Developed the Express.js backend with JSON APIs and JWT authentication for live value retrieval and portfolio tracking.

Organizations

Boy Scouts of America, Troop 222 — Huntersville, NC
Eagle Scout; Senior Patrol Leader and Assistant Scoutmaster — *2015 – Present*

- Mentor 30+ Scouts as Assistant Scoutmaster and lead outdoor programs focused on leadership, teamwork, and problem-solving.

Skills

Controls and Automation: Omron PLC programming (CX-Programmer, ladder logic), HMI development (DOPSoft), machine vision (Keyence IV3/IV4, IV Smart Navigator), stepper motion control, sensor integration (proximity, laser), solenoid and pneumatic actuation, robot control software (WPILib)
Electrical Design: Control panel design and fabrication, wiring and terminations, schematic capture (TinyCAD)
Mechanical Design: SolidWorks (Essentials for Part Design certificate), 3D printing (Prusa Slicer, Cura)
Languages: Java, Python, C/C++, Dart, SQL, JavaScript, HTML/CSS
Developer Tools: Git, Docker, VS Code, IntelliJ, TensorFlow, Google Cloud Platform, Android Studio, Xcode